MSA680 - Lab 2

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1 Introduction

In this lab, we are given data on growth of 27 participants (11 girls, 16 boys). Measurements of distance (mm) from the centre of the pituitary gland to the maxillary fissure were taken at the ages 8, 10, 12 and 14 years for the participants. Compared with the complete dataset (CC), two subsets of the complete data were also used – the first subset by Little and Rubin (AX) has 27 participants, but has some measurements at certain ages (e.g. participant 3's age 10 measurement) removed; the second subset (G2) has complete data for 18 participants (7 girls, 11 boys) only, where the 9 participants removed were the ones who had partial measurements erased in the AX dataset. A first look at the three datasets can be seen below.

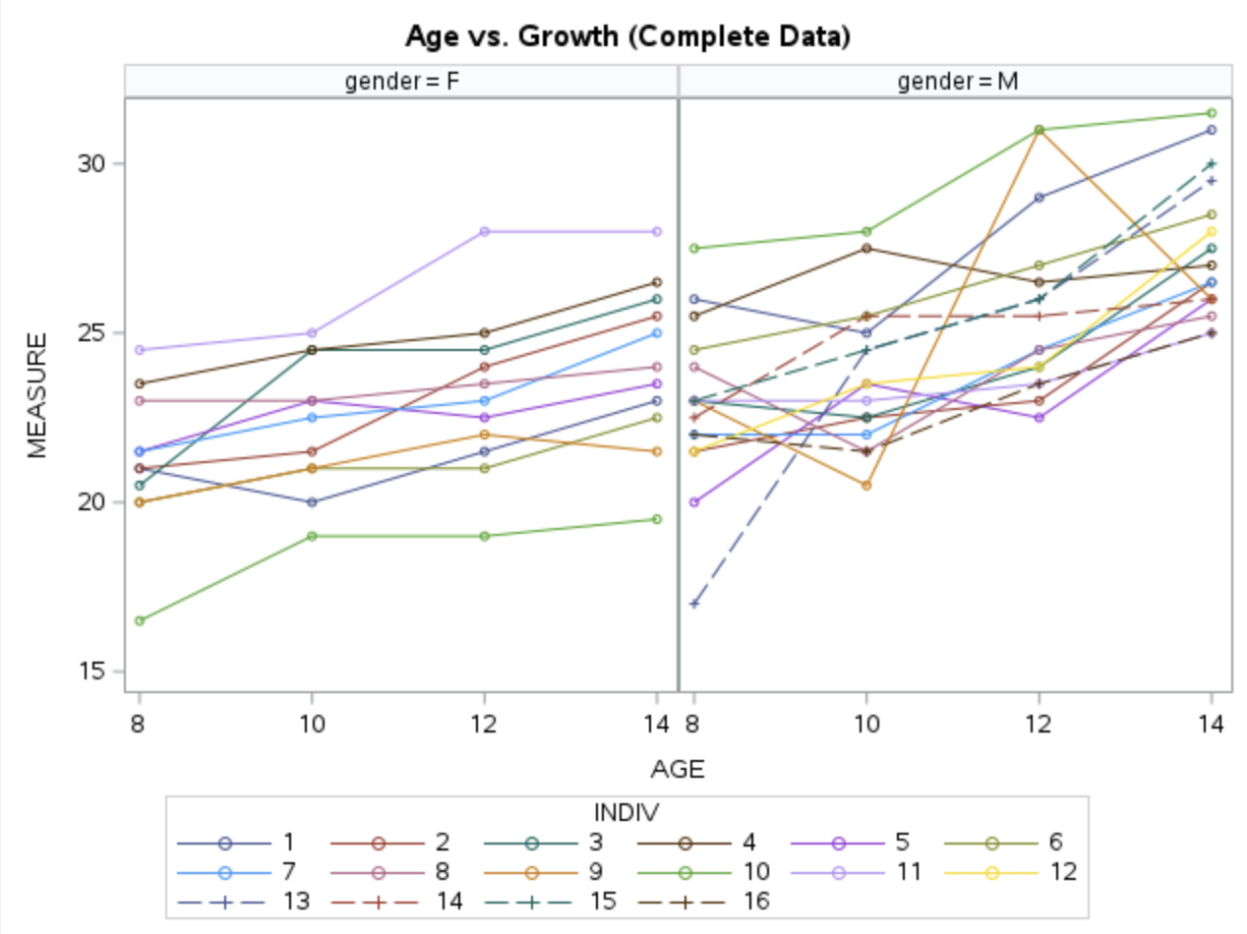


Figure 1: Age vs. Growth, Complete data (CC)

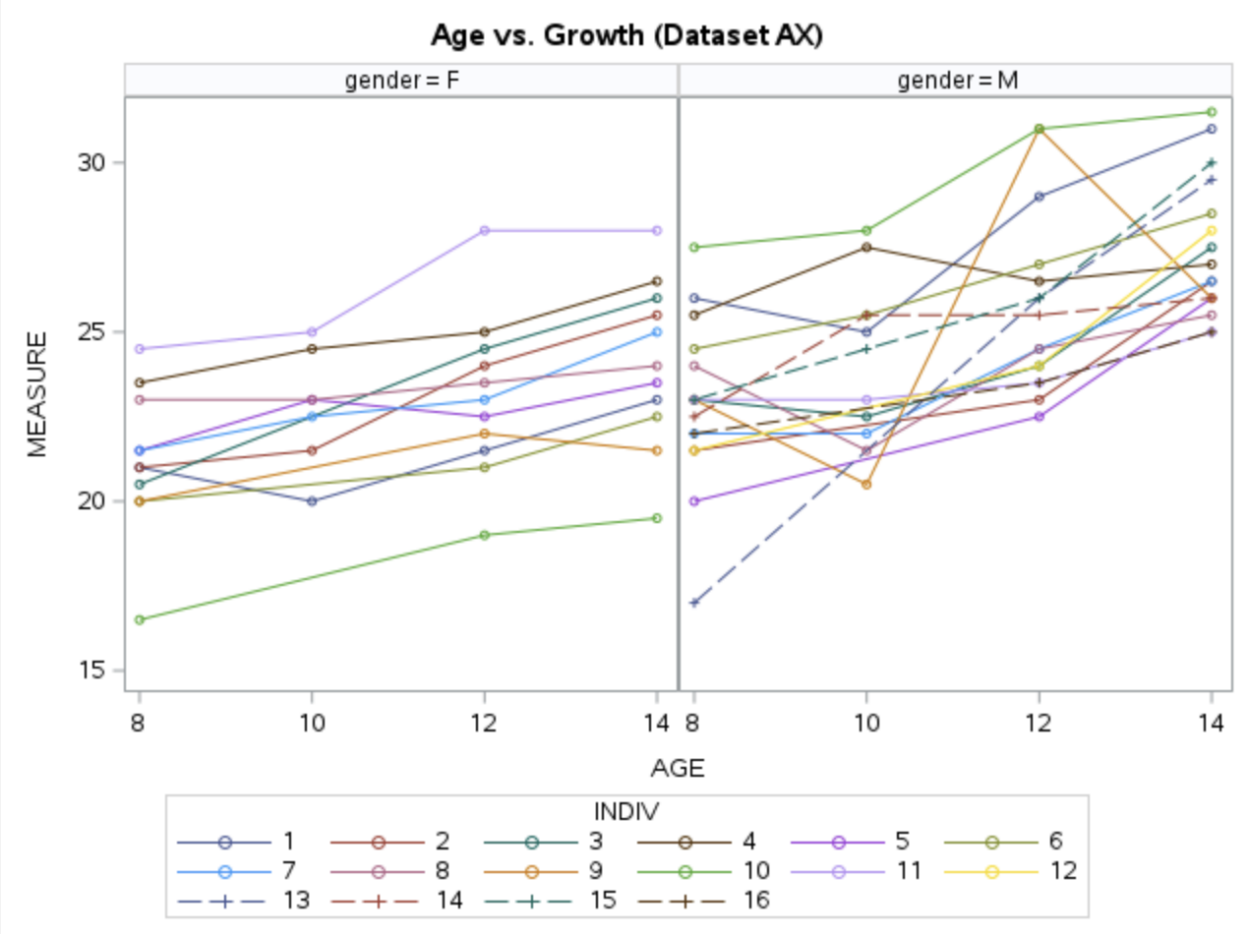


Figure 2: Age vs. Growth, Data Removed by Little and Rubin (AX)

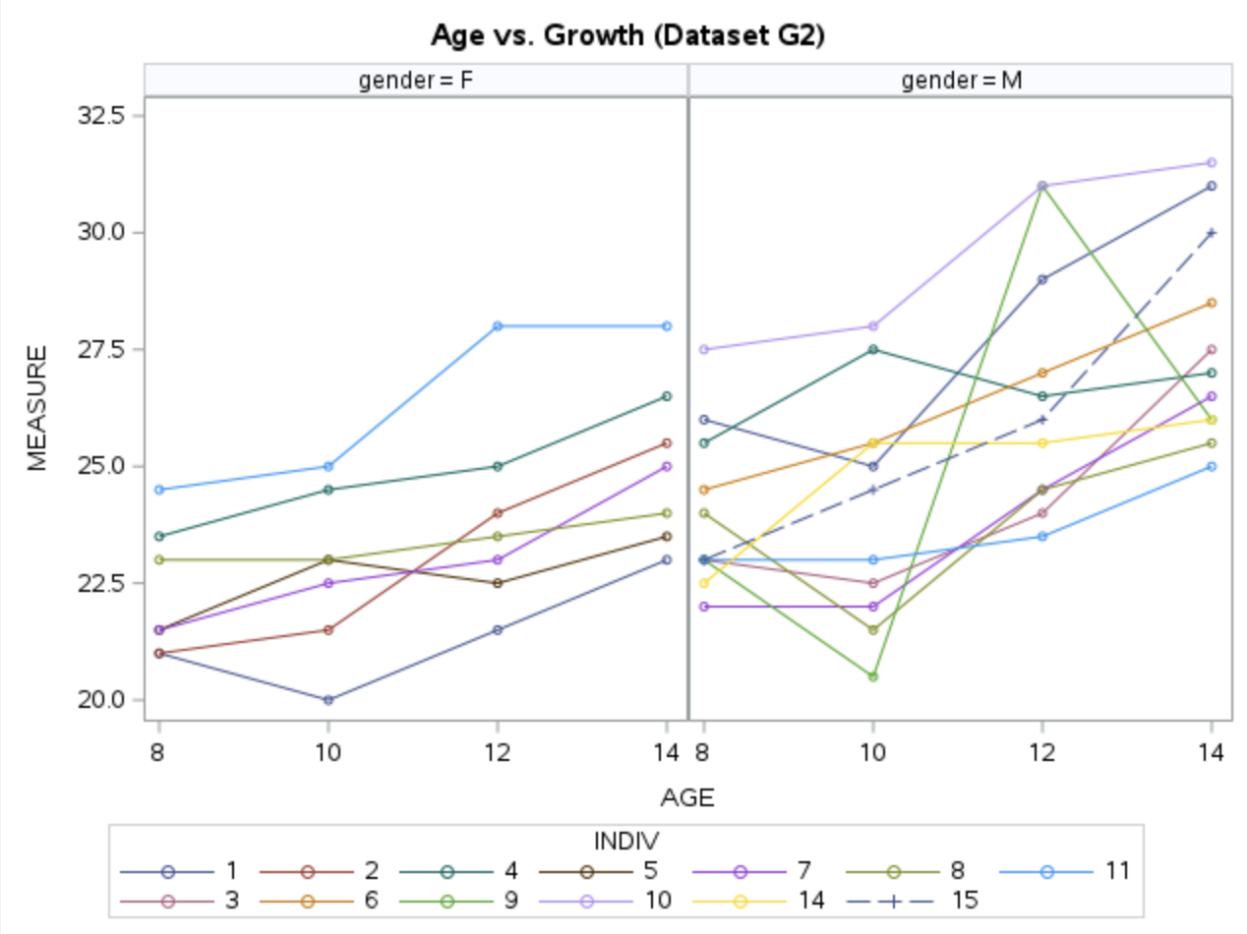


Figure 3: Age vs. Growth, Data with Missing Individual Values Completely Removed (G2)

Looking at the three datasets, we can see that in both boys and girls, there is a positive relationship between age and distance measured (mm). This is to be expected, as children between 8-14 generally do grow taller as they get older. There is much variation between individuals, and it seems that boys in general have lengthier distances measured at the same age compared with girls. There seem to be some odd trends going on within some subjects (e.g. boy participant #9 seems to have decreased distance at age 10, much increased distance at age 12, then decreased distance at age 14 again) – perhaps this is attributable to some measurement error.

2 Modelling (Assuming $\Sigma_i = \sigma_i^2 I$)

The modeling in this lab was done in SAS (see code in appendix). A reparametrized age parameter, `age2`=$\frac{\text{age}-11}{3}$, was added to the data. This transforms and centers the covariate to be between -1 (corresponding to 8 years of age) and 1 (corresponding to 14 years of age). This also helps with interpretation, as the mean transformed age of `age2`=0 corresponds to the intercept, which is the "average" growth of a subject at the middle of the study.

The mixed effects model was defined as below:

$$Y_{ij} = \beta_0 + \beta_{01}x_i + \beta_{10}(1 - x_i)t_{ij} + \beta_{11}x_i t_{ij} + b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij} \tag{1}$$

where the index i represents the participant, and $j \in \{1, 2, 3, 4\}$ represents the time the measurement was taken. Furthermore, we have the following variables:

i : Index of Participant

j : Index of time the measurement was taken

Y_{ij} : Distance measured (mm) for participant i at time index j

x_i : Binary categorical variable for participant i; boy (0) or girl (1)

t_{ij} : Age for participant i at time index j

By default, the conditional covariance matrix Σ is diagonal with the same variance, i.e. $\sigma^2 I$ due to the errors ε being assumed to be independent from each other and having constant variance. To interpret the other parameters, we can first rewrite equation 1 as follows:

$$
\begin{aligned}
Y_{ij} &= \beta_0 + \beta_{01}x_i + \beta_{10}(1-x_i)t_{ij} + \beta_{11}x_i t_{ij} + b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij} \\
&= \begin{cases} \beta_0 + \beta_{10}t_{ij} + b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij}, & \text{for } x_i = 0 \\ \beta_0 + \beta_{01} + \beta_{11}t_{ij} + b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij}, & \text{for } x_i = 1 \end{cases} \\
&= \begin{cases} (\beta_0 + b_{0i}) + t_{ij}(\beta_{10} + b_{1i}) + \varepsilon_{ij}, & \text{for } x_i = 0 \\ (\beta_0 + \beta_{01} + b_{0i}) + t_{ij}(\beta_{11} + b_{1i}) + \varepsilon_{ij}, & \text{for } x_i = 1 \end{cases}
\end{aligned}
$$

In other words, this is an unequal slopes linear mixed effects model with age and sex as covariates. The β's represent the parameters for the fixed effects, while the b's represent the parameters for the random/individual effects. For the boys ($x_i = 0$), the intercept $\beta_0 + b_{01}$ is contributed by the fixed effects β_0 as well as the individual effects b_{01}; the slope $\beta_{10} + b_{1i}$ is contributed by the fixed effects β_{10} as well as the individual effects b_{1i}. For the girls ($x_i = 1$) the intercept $\beta_0 + \beta_{01} + b_{0i}$ is contributed by the fixed effects $\beta_0 + \beta_{01}$ (compared to the boys' baseline) as well as the individual effects b_{0i}; the slope $\beta_{11} + b_{1i}$ is contributed by the fixed effects β_{11} (different from the boys'), as well as the individual effects b_{1i}.

For the complete dataset, each participant has four measurements ($j \in \{1, 2, 3, 4\}$) at ages 8, 10, 12, and 14. The design matrices X_i for the untransformed age and Z_i are thus as follows:

$$
\boldsymbol{X_i} = \begin{bmatrix} 1 & x_i & 1-x_i & 8 \\ 1 & x_i & 1-x_i & 10 \\ 1 & x_i & 1-x_i & 12 \\ 1 & x_i & 1-x_i & 14 \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_{01} \\ \beta_{10} \\ \beta_{11} \end{bmatrix}
$$

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		16.3406	1.0158	25	16.09	<.0001
AGE		0.7844	0.08574	25	9.15	<.0001
gender	F	1.0685	1.5914	54	0.67	0.5048
gender	M	0	.	.	.	.
AGE*gender	F	-0.3071	0.1343	54	-2.29	0.0262
AGE*gender	M	0	.	.	.	.

(a) Fixed Effects Estimates (untransformed age)

Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1	5.5484	-0.2681
2	AGE	1	-0.2681	0.03063

(b) D matrix (untransformed age)

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		24.9688	0.4867	25	51.30	<.0001
age2		2.3531	0.2572	25	9.15	<.0001
gender	F	-2.3097	0.7626	54	-3.03	0.0038
gender	M	0	.	.	.	.
age2*gender	F	-0.9213	0.4030	54	-2.29	0.0262
age2*gender	M	0	.	.	.	.

(c) Fixed Effects Estimates (transformed age)

Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1	3.3558	0.2063
2	age2	1	0.2063	0.2756

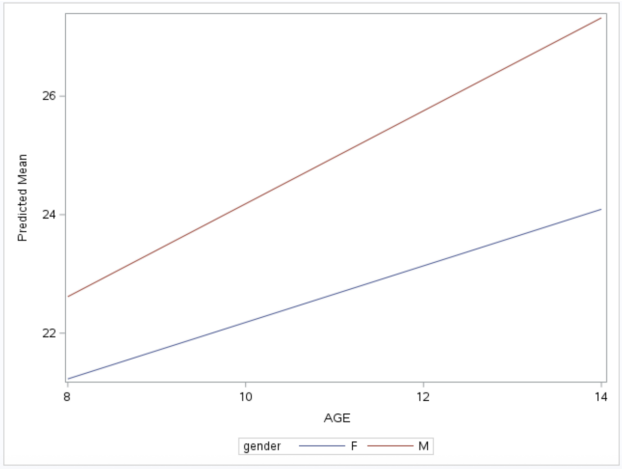
(d) D matrix (transformed age)

Figure 4: Age vs. Growth models parameter estimates for the complete dataset (CC)

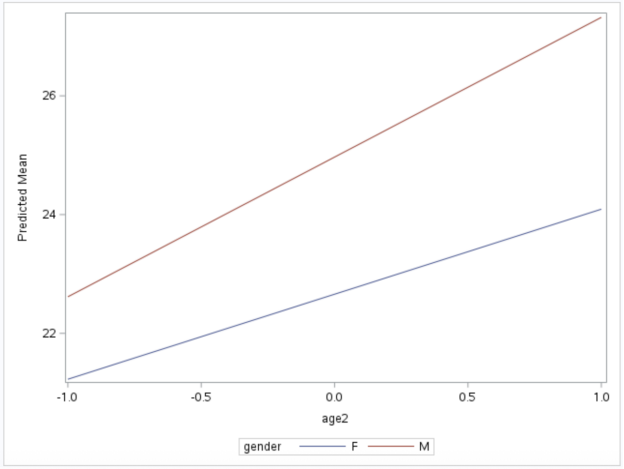
$$\boldsymbol{Z_i} = \begin{bmatrix} 1 & 8 \\ 1 & 10 \\ 1 & 12 \\ 1 & 14 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} b_{0i} \\ b_{1i} \end{bmatrix}$$

3 Results (Model assuming $\Sigma_i = \sigma_i^2 I$)

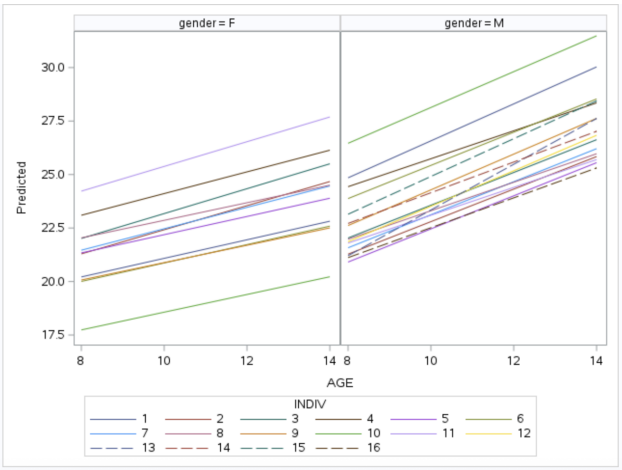
Results will be presented first, and interpretations will follow in the next section.



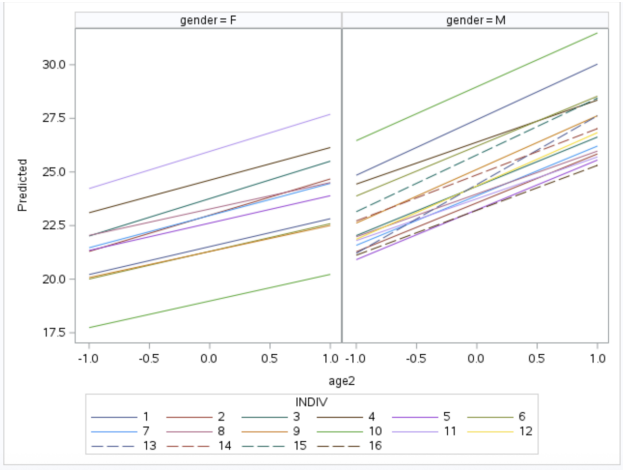
(a) Mean Responses (untransformed age)



(b) Mean Responses (transformed age)



(c) Predicted Responses (untransformed age)



(d) Predicted Responses (transformed age)

Figure 5: Age vs. Growth model graphs for the complete dataset (CC)

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		16.2658	1.1169	25	14.56	<.0001
AGE		0.7891	0.09163	25	8.61	<.0001
gender	F	0.9382	1.7519	45	0.54	0.5949
gender	M	0	.	.	.	.
AGE*gender	F	-0.2990	0.1437	45	-2.08	0.0431
AGE*gender	M	0	.	.	.	.

(a) Fixed Effects Estimates (untransformed age)

Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1	8.3550	-0.4653
2	AGE	1	-0.4653	0.04415

(b) D matrix (untransformed age)

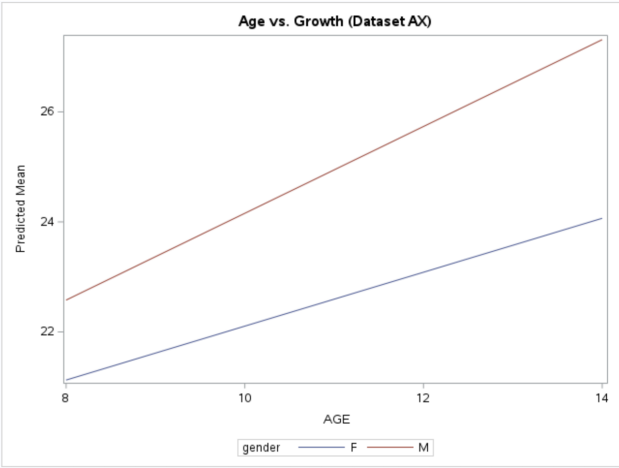
Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		24.9454	0.4968	25	50.21	<.0001
age2		2.3672	0.2749	25	8.61	<.0001
gender	F	-2.3504	0.7788	45	-3.02	0.0042
gender	M	0	.	.	.	.
age2*gender	F	-0.8969	0.4310	45	-2.08	0.0431
age2*gender	M	0	.	.	.	.

(c) Fixed Effects Estimates (transformed age)

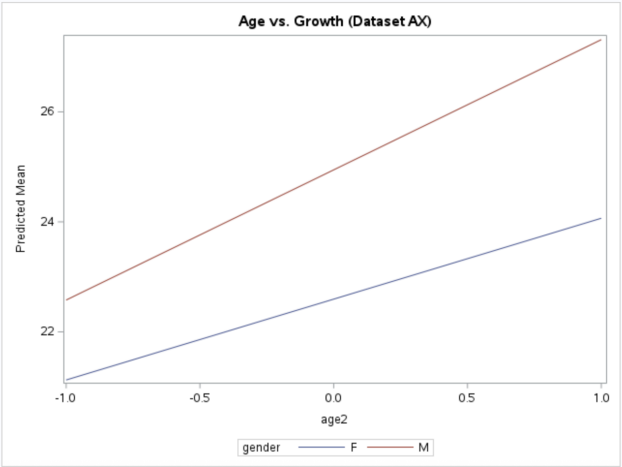
Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1	3.4615	0.06117
2	age2	1	0.06117	0.3973

(d) D matrix (transformed age)

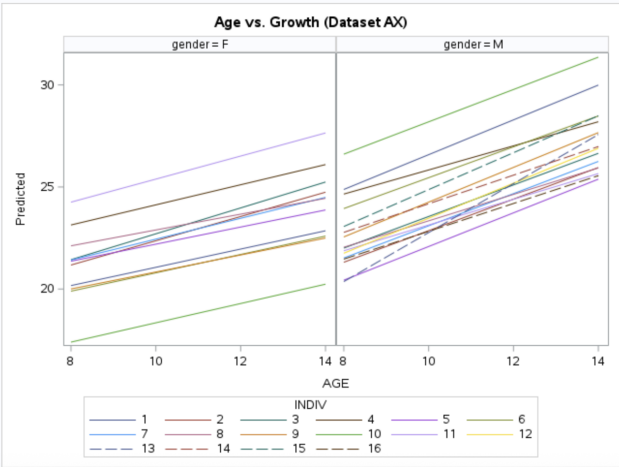
Figure 6: Age vs. Growth models parameter estimates for dataset with select observations removed (AX)



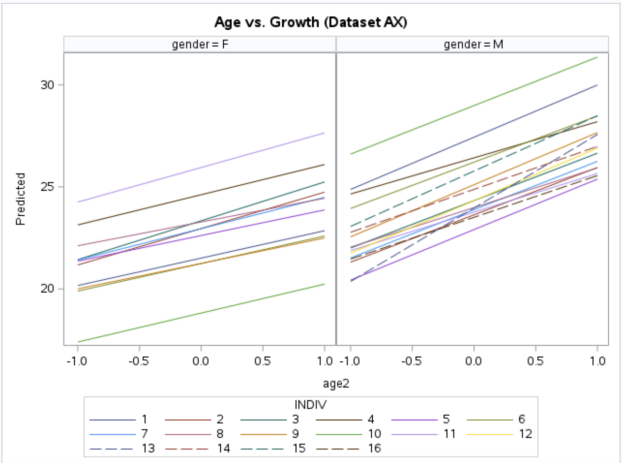
(a) Mean Responses (untransformed age)



(b) Mean Responses (transformed age)



(c) Predicted Responses (untransformed age)



(d) Predicted Responses (transformed age)

Figure 7: Age vs. Growth model graphs for dataset with select observations removed (AX)

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		18.1773	1.0154	16	17.90	<.0001
AGE		0.6750	0.09046	16	7.46	<.0001
gender	F	0.1156	1.6283	36	0.07	0.9438
gender	M	0	.	.	.	.
AGE*gender	F	-0.2000	0.1451	36	-1.38	0.1765
AGE*gender	M	0	.	.	.	.

(a) Fixed Effects Estimates (untransformed age)

Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1		0.1252
2	AGE	1	0.1252	

(b) D matrix (untransformed age)

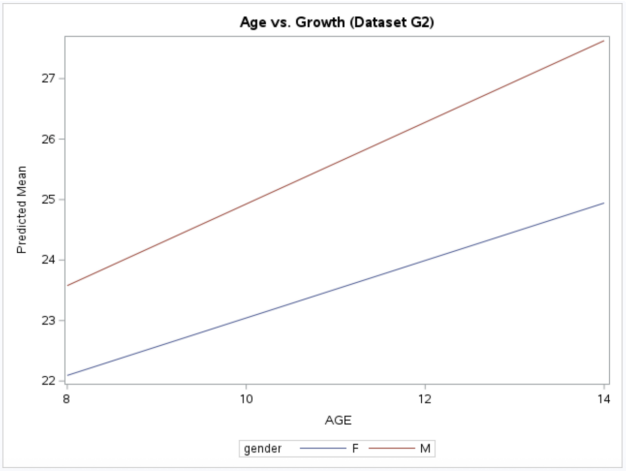
Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		25.6023	0.5401	16	47.41	<.0001
age2		2.0250	0.2729	16	7.42	<.0001
gender	F	-2.0844	0.8660	36	-2.41	0.0213
gender	M	0	.	.	.	.
age2*gender	F	-0.6000	0.4376	36	-1.37	0.1789
age2*gender	M	0	.	.	.	.

(c) Fixed Effects Estimates (transformed age)

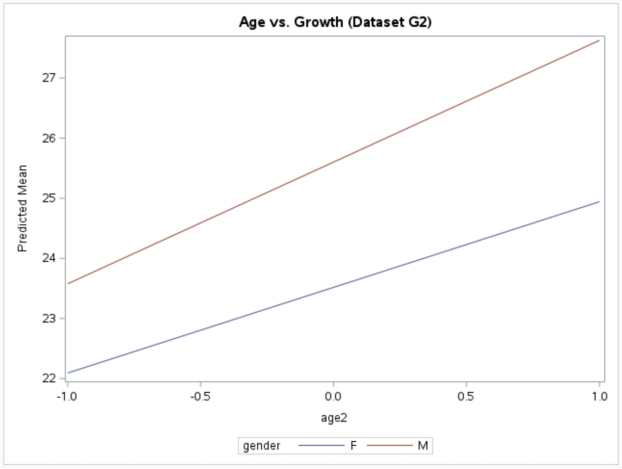
Estimated G Matrix				
Row	Effect	IDNR	Col1	Col2
1	Intercept	1	2.7531	0.4984
2	age2	1	0.4984	

(d) D matrix (transformed age)

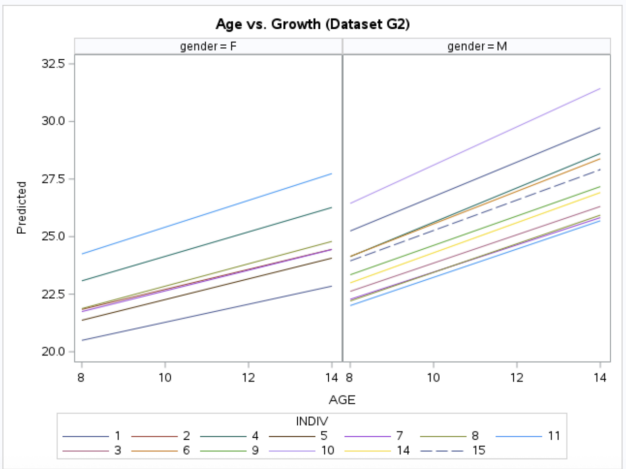
Figure 8: Age vs. Growth models parameter estimates for dataset with individuals with missing observations removed (G2)



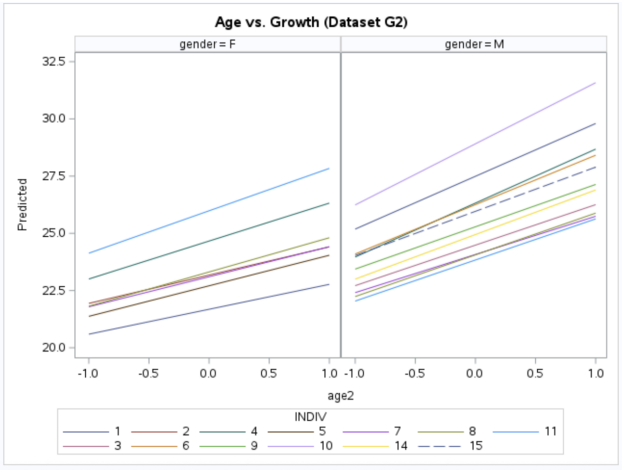
(a) Mean Responses (untransformed age)



(b) Mean Responses (transformed age)



(c) Predicted Responses (untransformed age)



(d) Predicted Responses (transformed age)

Figure 9: Age vs. Growth model graphs for dataset with individuals with missing observations removed (G2)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.7398			
2		1.7398		
3			1.7398	
4				1.7398

(a) Σ_1 matrix for CC data (untransformed age)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.7398			
2		1.7398		
3			1.7398	
4				1.7398

(b) Σ_1 matrix for CC data (transformed age)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.7665			
2		1.7665		
3			1.7665	
4				1.7665

(c) Σ_1 matrix for AX data (untransformed age)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.7665			
2		1.7665		
3			1.7665	
4				1.7665

(d) Σ_1 matrix for AX data (transformed age)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.8003			
2		1.8003		
3			1.8003	
4				1.8003

(e) Σ_1 matrix for G2 data (untransformed age)

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.8207			
2		1.8207		
3			1.8207	
4				1.8207

(f) Σ_1 matrix for G2 data (transformed age)

Figure 10: $\Sigma_1 = \sigma^2 I$ matrix for individual 1 in all models under constant variance assumption

4 Interpretations (Model assuming $\Sigma_i = \sigma_i^2 I$)

We can first notice that looking visually at the graphs of the mean responses and all predicted responses, they look near identical for untransformed age vs. transformed age – the only difference being the "center" of the age measurements shifted to 0 for the transformed age. This is especially reflected in the slope parameter estimates, where $\hat{\beta}_{10} = 0.7844$ (untransformed age) and $\hat{\beta}_{10} = 2.3531$ (transformed age), which is exactly different by a factor of 3 – just as expected from the age transformation since we divided by 3. For ease of interpretability, the following interpretations and models will be done on the transformed age.

First, we can compare the complete dataset (CC) modeled with untransformed age and transformed age as seen in Figures 4 and 5 above. At the $\alpha = 0.05$ level, all β fixed effects parameters were significant for the complete data model with transformed age and sex as covariates. The intercept $\beta_0 = 24.9688$ represents the parameter estimate of the measured distance (mm) for boys at the age of 11 years old. For girls, the estimated measured distance (mm) at age 11 is $\beta_0 + \beta_{01} = 24.9688 - 2.3097 = 22.6591$, significantly lower compared with the boys. Regarding the effect of age, an increase in 1 year of age ($\frac{1}{3}$ years transformed age) for boys results in a $\beta_{10} = 0.7844$mm increase in measured distance, whereas for girls an increase of 1 year of age results in a $\beta_{11} = 0.7844 - 0.3071 = 0.4773$mm increase of measured distance. In other words, girls between the age of 8-14 years old grow on average at a slower rate than boys of the same age.

Performing the same analysis on the AX dataset (select observations removed), the results can be seen in Figures 6 and 7 below. Graphically, both the predicted individual growth profiles and the group mean predicted growth profiles look very similar to the complete dataset. Looking at the parameter estimates for both the untransformed and transformed ages for the AX dataset, they are very similar to the parameter estimates for the CC dataset, and have the same levels of significance at the $\alpha = 0.05$ level.

Finally, performing the same analysis on the G2 dataset (individuals with missing ob-

servations removed), the results can be seen in Figures 8 and 9 below. The results for this dataset are quite different from the previous two. Firstly, we can notice that the graphs for the predicted responses are much more similar comparing boys to girls – the slopes look more similar, meaning that the growth rate among the boys and girls is less different. Indeed, the difference in parameter estimates for the girls' intercept and slope compared to the baseline parameter estimates for the boys' intercept and slope were not as different compared to the previous models. Significance level wise, the interaction term for slope was not significant at the $\alpha = 0.05$ level, meaning that the slopes might not be significantly different between girls and boys. Furthermore, the estimated D matrix was not positive definite for both the untransformed age and the transformed age. This is an issue, as variances for parameters were estimated to be negative, which is not realistic. Removing the NOBOUND restriction in SAS allowed for the diagonal elements in the D matrix to be non-positive, however much interpretability was lost as, again, the variance of age/intercept cannot be negative in real life.

Looking at the D matrices (G matrix in R) and Σ_i matrices (R matrix in R), we can also see some patterns across the models for the three different datasets. Focusing on the D matrix for untransformed age as seen in Figures 4(b), 6(b), and 8(b), we can see that the variability of the group level random effects are on the main diagonal. As we decrease the amount of data, the variation increases, which is to be expected (e.g. for intercept, the variation goes from 5.55(CC) to 8.3550 (AX)). Furthermore, for the dataset with fewest data (G2), the D matrix was unable to be calculated due to the negative variances. For the D matrix for transformed age as seen in Figures 4(c), 6(c), and 8(c) we see a similar pattern on the main diagonals, however we see that the covariance between age and intercept is now positive, due to the age transformation. Focusing on the individual level effects for individual 1 (i.e. Σ_1) in Figure 10, we can see that firstly the matrix is assumed to be diagonal with the same variance across different measurements. In other words, the same variance is assumed for subsequent measurements, and measurements are not assumed to affect each other (off

diagonal terms are all zero). We can see that it is a 4x4 matrix (since individual 1 has 4 measurements), and there is not much difference between untransformed/transformed age, and also not a big difference across the three datasets; this is to be expected, since these are individual level effects.

5 Modelling (Assuming AR(1) structure Σ_i)

In this section, we will fit another mixed effects model. However, instead of following the constant variance assumption of $\Sigma_i = \sigma_i^2$, we will allow a AR(1) serial structure (and AR(1) structure with measurement error) for the Σ_i matrix. In other words, while the variance of the diagonal elements are assumed to be constant σ_i^2, the measurements "close" in terms of time proximity will have a higher covariance ρ_i^k, where k is the "distance" of measurements removed from the current measurement. For a participant i who had four growth measurements, this would be a 4x4 matrix with the following structure:

$$\Sigma_i = \sigma_i^2 \begin{bmatrix} 1 & \rho_i & \rho_i^2 & \rho_i^3 \\ \rho_i & 1 & \rho_i & \rho_i^2 \\ \rho_i^2 & \rho_i & 1 & \rho_i \\ \rho_i^3 & \rho_i^2 & \rho_i & 1 \end{bmatrix}$$

Furthermore, in this new model we will only consider random intercepts, and no random slopes. This means that at the individual level, the rate of growth at is assumed to be the same as the group level with no variation, and thus there will be no more b_{1i} term as in the previous model. In other words:

$$
\begin{aligned}
Y_{ij} &= \beta_0 + \beta_{01}x_i + \beta_{10}(1-x_i)t_{ij} + \beta_{11}x_i t_{ij} + b_{0i} + \varepsilon_{ij} \\
&= \begin{cases} \beta_0 + \beta_{10}t_{ij} + b_{0i} + \varepsilon_{ij}, & \text{for } x_i = 0 \\ \beta_0 + \beta_{01} + \beta_{11}t_{ij} + \varepsilon_{ij}, & \text{for } x_i = 1 \end{cases} \\
&= \begin{cases} (\beta_0 + b_{0i}) + \beta_{10}t_{ij} + \varepsilon_{ij}, & \text{for } x_i = 0 \\ (\beta_0 + \beta_{01} + b_{0i}) + \beta_{11}t_{ij} + \varepsilon_{ij}, & \text{for } x_i = 1 \end{cases}
\end{aligned}
$$

For simplicity's sake, in this model we will only focus on the complete dataset CC, and use the transformed age as a covariate instead of untransformed age. Again, the results will be presented below first, then interpretations after. The results for the model assuming AR(1) structure Σ_i) is given in Figure 11, and the results for the model assuming AR(1) + measurement error structure Σ_i is given in Figure 12.

6 Results (Model assuming AR(1) structure Σ_i)

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		24.9640	0.4869	25	51.27	<.0001
age2		2.3572	0.2274	79	10.37	<.0001
gender	F	-2.3044	0.7628	79	-3.02	0.0034
gender	M	0	.	.	.	.
age2*gender	F	-0.9270	0.3563	79	-2.60	0.0111
age2*gender	M	0	.	.	.	.

(a) Fixed Effects Estimates

Estimated G Matrix			
Row	Effect	IDNR	Col1
1	Intercept	1	3.3543

(b) D Matrix

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.8873	-0.08991	0.004283	-0.00020
2	-0.08991	1.8873	-0.08991	0.004283
3	0.004283	-0.08991	1.8873	-0.08991
4	-0.00020	0.004283	-0.08991	1.8873

(c) Σ_1 Matrix

Figure 11: Transformed Age vs. Growth model parameter estimates for complete dataset (CC) with AR(1) structure Σ_i variance-covariance matrices

Solution for Fixed Effects						
Effect	gender	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		24.9688	0.4867	25	51.30	<.0001
age2		2.3790	0.2221	79	10.71	<.0001
gender	F	-2.3097	0.7626	79	-3.03	0.0033
gender	M	0	.	.	.	.
age2*gender	F	-0.9577	0.3480	79	-2.75	0.0073
age2*gender	M	0	.	.	.	.

(a) Fixed Effects Estimates

Estimated G Matrix			
Row	Effect	IDNR	Col1
1	Intercept	1	3.3834

(b) D Matrix

Estimated R Matrix for IDNR 1				
Row	Col1	Col2	Col3	Col4
1	1.8544	-0.2255	0.2255	-0.2255
2	-0.2255	1.8544	-0.2255	0.2255
3	0.2255	-0.2255	1.8544	-0.2255
4	-0.2255	0.2255	-0.2255	1.8544

(c) Σ_1 Matrix

Figure 12: Transformed Age vs. Growth model parameter estimates for complete dataset (CC) with AR(1) + Measurement error structure Σ_i variance-covariance matrices

7 Interpretation (Model assuming AR(1) structure Σ_i

We can compare the results for the AR(1) model (transformed age, CC dataset) in Figure 11 with the results from the model assuming $\Sigma_i = \sigma_i^2 I$ (transformed age, CC dataset) in Figures 4c, 4d and 10b. Firstly for the fixed effects, we can notice that for all parameter estimates there is extremely little difference between the two models (e.g. slope of 2.3532 vs. 2.3572). For the individual effects, the G matrix for the new model only has one entry due to the removal of the age random effect. Comparing variances for intercept, it is again very similar (3.3558 vs. 3.3543). This is an indication that perhaps the inclusion of random slopes is not necessary.

Furthermore, if we focus on the Σ_1 matrix for the individual-level effects for the 1st individual, we can see that the main diagonal variances are again very similar (1.7398 vs. 1.8873). In the AR(1) structure matrix, the off diagonal entries are all extremely close to 0, indicating a very small ρ value and therefore little relationship between previous observations on subsequent observations in the individual time series data. This indicates that it is likely that the change to AR(1) serial structure Σ_i matrix was not different from the simpler $\Sigma_i = \sigma_i^2 I$ matrix model.

Finally, comparing the AR(1) model (transformed age, CC dataset) in Figure 11 with AR(1) model with observation error component (transformed age, CC dataset) in Figure 12, we again do not see significant differences in fixed effects parameter estimates, variances in the G matrix, or variance-covariance matrix Σ_1 for individual level effects for individual 1. The main difference is the structure of the Σ_1 matrix, whose off-diagonal terms showed slightly higher covariance.

These AR(1) models are therefore not necessarily more suitable to the data than previous models. This is because the results are quite familiar, and secondly the off diagonal terms were extremely close to zero in the AR(1) models. A random effects model would be useful if there is a long-term"causational" effect expected (e.g. depending on one's genetics, growth rate can be faster/slower), whereas for serial correlation model would be useful if there are

short-term events that might affect near-neighbors (e.g. recessions or a stock market crash will affect prices close in time).

The unbalance in data (more boys than girls, also some missing measurements in the AX dataset) had two minor effects. Firstly the standard errors in the fixed effects models for β_{01} (girls' group level intercept compared with boys' baseline) and β_{11} (girls' group level slope compared with boys' baseline) were high due to the small sample size, thus we could not say if some of the parameter estimates were significantly different from zero. Secondly, for the G2 dataset with many individuals removed, the slopes for the boys and girls were not significantly different, and the D matrices were not invertible. Perhaps more balanced data with equal group sizes, or a larger sample size in general, would allow for these drawbacks to be fixed.

8 Appendix - SAS Code

```
/* Setting path and library */
libname growth "/home/u64435675/sasuser.v94/Labs";

/* Reading in the data*/
data growthcc;
set growth.growthcc;
run;

data growthax;
set growth.growthax;
run;

data growth2;
```

```
set growth.growth2;
run;

/* Refactoring sex, adding transformed age, then clean up datasets*/
data growth_cleaned;
    set growthcc; /* CHANGE HERE to look at models for the other datasets */
    age2 = (age - 11) / 3;
    if sex = 1 then gender = "M";
    else gender = "F";
    agec = age; /* Assigning as categorical for REPEATED statement later */
    age2c = age2; /* Assigning as categorical for REPEATED statement later */
run;

/* Visualizing the data*/
proc sgpanel data = growth_cleaned;
panelby gender / columns = 2;
series x = age y = measure / group = indiv markers ;
/* To change title based on dataset*/
title 'Age vs. Growth (Dataset CC)';
run;

/* Fitting the model (untransformed age)*/
proc mixed data = growth_cleaned; /* add NOBOUND option here */
class idnr indiv sex gender agec age2c;
model measure = age gender age*gender/ solution chisq OUTP = pred outpm = ave;
random intercept age / type=un subject=idnr G Gcorr;
repeated agec / subject=idnr R Rcorr;
```

```
run;

/* Visualizing the predictions*/
proc sort data = pred;
by age;
run;

proc sgplot data = ave;
series y = pred x = age / group = gender;
run;

proc sgpanel data = pred;
panelby gender;
series y = pred x = age / GROUP = indiv;
run;

/* Fitting the model (transformed age)*/
proc mixed data = growth_cleaned; /* add NOBOUND option here */
class idnr indiv sex gender agec age2c;
model measure = age2 gender age2*gender/ solution chisq OUTP = pred outpm = ave;
random intercept age2 / type=un subject=idnr G Gcorr;
repeated age2c / subject=idnr R Rcorr;
run;

/* Visualizing the predictions*/
proc sort data = pred;
by age2;
```

```
run;


proc sgplot data = ave;
series y = pred x = age2 / group = gender;
run;


proc sgpanel data = pred;
panelby gender;
series y = pred x = age2 / GROUP = indiv;
run;


/* Fitting random-int only model to AR(1) variance-covariance matrix (untransformed age)
proc mixed data = growth_cleaned;
class idnr indiv sex gender agec age2c;
model measure = age2 gender age2*gender/ solution chisq OUTP = pred outpm = ave;
random intercept / type=un subject=idnr G Gcorr;
repeated age2c /type=AR(1) subject=idnr R Rcorr;
run;


/* Fitting random-int only model to AR(1) + measurement error variance-covariance matrix
proc mixed data = growth_cleaned;
class idnr indiv sex gender agec age2c;
model measure = age2 gender age2*gender/ solution chisq OUTP = pred outpm = ave;
random intercept / type=un subject=idnr G Gcorr;
repeated age2c /type=AR(1) local subject=idnr R Rcorr; /* LOCAL adds measurement error c
run;
```